

Percentage Increase

Jonathan R. Bacolod

A **percentage increase** describes how much a quantity grows compared to its original value.

Amount of Increase

To find the **amount of increase**, subtract the original value from the new value.

$$\text{Increase} = \text{New Value} - \text{Original Value}$$

Percent Increase

The **percent increase** is calculated using:

$$\text{Percent Increase} = \frac{\text{Increase}}{\text{Original Value}} \times 100\%$$

New Value

To find the new value after an increase:

$$\text{New Value} = \text{Original Value} + \text{Increase}$$

Finding the Increase

If the percentage increase is known:

$$\text{Increase} = \text{Original Value} \times \left(\frac{\text{Percent}}{100} \right)$$

Examples

Example 1

A notebook costs ₱80. Its price increased to ₱100. Find the percentage increase.

$$\text{Increase} = 100 - 80 = 20$$

$$\text{Percent Increase} = \frac{20}{80} \times 100\% = 25\%$$

Example 2

A bicycle worth ₱4,500 increased in price by 20%. What is its new price?

$$\text{Increase} = 4500(0.20) = \text{₱}900$$

$$\text{New Price} = 4500 + 900 = \text{₱}5,400$$

Example 3

A school's enrollment increased from 800 students to 920 students. Find the percentage increase.

$$\text{Increase} = 920 - 800 = 120$$

$$\text{Percent Increase} = \frac{120}{800} \times 100\% = 15\%$$

Practice Exercises

Practice 1

The population of a barangay increased from 2,500 to 3,000 residents. Find the percentage increase.

Practice 2

Maria saved ₱1,200 in January and ₱1,500 in February. By what percentage did her savings increase?

Practice 3

A farmer harvested 600 kilograms of rice this year compared to 500 kilograms last year. Find the percentage increase.

Practice 4

A worker's monthly salary of ₱18,000 increased by 15%. What is the new salary?

Practice 5

A water tank contained 240 liters of water. After rainfall, the amount increased to 300 liters. Find the percentage increase.

Practice 6

A store sold 150 bags in one week and 195 bags the following week. Find the percentage increase in sales.

Practice 7

The value of a gadget increased from ₱12,000 to ₱14,400. Find the percentage increase.

Activities

Activity 1

A pen costs ₱60. Its price increased to ₱75.
Find the percentage increase.

Activity 2

The population of a town increased from 4,000 to 4,800 residents. Find the percentage increase.

Activity 3

Carlo saved ₱2,000 in March and ₱2,500 in April. By what percentage did his savings increase?

Activity 4

A laptop worth ₱25,000 increased in price by 12%. What is its new price?

Activity 5

A school's enrollment increased from 1,200 students to 1,380 students. Find the percentage increase.

Activity 6

A farmer harvested 900 kilograms of corn this year compared to 750 kilograms last year. Find the percentage increase.

Activity 7

A worker's monthly salary of ₱22,000 increased by 10%. What is the new salary?

Activity 8

A water tank contained 350 liters of water. After rainfall, the amount increased to 420 liters. Find the percentage increase.

Activity 9

A store sold 240 shirts in one week and 300 shirts the following week. Find the percentage increase in sales.

Activity 10

The value of a tablet increased from ₱15,000 to ₱18,000. Find the percentage increase.

Thank you!